

UNIT 1

MEASUREMENT OF PHYSICAL QUANTITIES

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- Concept of
 - (i) significant digits
 - (ii) accuracy versus precision
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- Least-count of an apparatus
- Methods to measure least-count with specific examples of travelling microscope and spectrometer

Physical Measurements



Quantities Tools Units



Mass
Length
Time



Kg
m
s



PHYSICAL QUANTITIES

- A physical quantity is quantity which can be measured.

For Example: length, mass, time, temperature, electric current, light intensity, volume and many others.

- A physical quantity is always measured of natural non-living objects.
- Physical quantities have numerical magnitude and unit.

For Example: If the length of a student is 104cm, then 104 is its numerical magnitude and cm is the unit of measurement.

Physical Quantities

```
graph TD; A[Physical Quantities] --> B[Fundamental physical quantities]; A --> C[Derived physical quantities]; B --- D[Cannot be defined in terms of other physical quantities]; B --- E[e.g. Length<br/>Time<br/>Mass]; C --- F[Can be defined in terms of Fundamental physical quantities]; C --- G[e.g. Volume<br/>Speed<br/>Acceleration];
```

Fundamental physical quantities

Cannot be defined in terms of other physical quantities

e.g. Length
Time
Mass

Derived physical quantities

Can be defined in terms of Fundamental physical quantities

e.g. Volume
Speed
Acceleration

SYSTEM OF UNITS

CGS System

Introduced in **France**

Gaussian system of units

cm for length

g for mass

s for time

MKS System

Introduced in **France**

French system of units

m for length

kg for mass

s for time

FPS System

Introduced in **Britain**

British system of units

ft for length

lb for mass

s for time

The 7 SI base units of Matter

(7 Fundamental quantities)

Fundamental quantities

$$\text{Speed} = \frac{\text{distance (m)}}{\text{time (s)}}$$

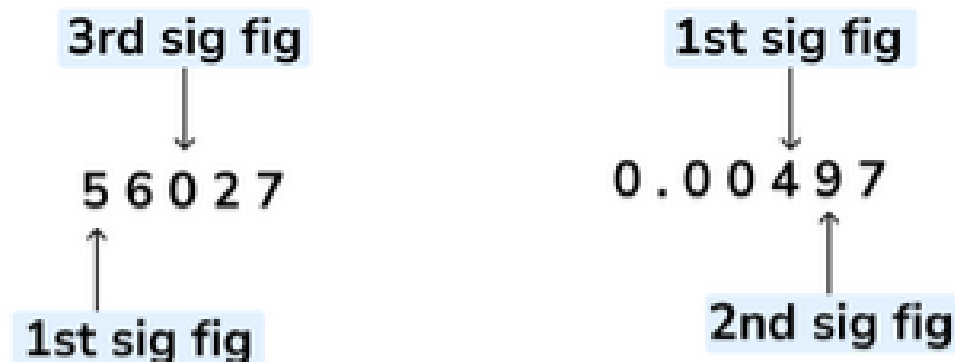
Derived quantity

- | | | | | |
|-----|---------------------|---|----------|------|
| ① • | Length | → | meter | (m) |
| ② • | Time | → | second | (s) |
| ③ • | Temperature | → | Kelvin | (K) |
| ④ • | Mass | → | kilogram | (kg) |
| ⑤ • | Amount of substance | → | Mole | (M) |
| ⑥ • | Electric current | → | Ampere | (A) |
| ⑦ • | Light intensity | → | candela | (cd) |

Significant figures

“Significant” means “important”. The first significant figure (or significant digit) of a number is the most important digit which expresses the size of the number; it is the first non-zero digit.

E.g.



Significant figures / significant digits

The significant figures of a given number are those significant or important digits, which convey the meaning according to its accuracy.

For example, 6.658 has four significant digits. These substantial figures provide precision to the numbers.

Rules for Significant Figures

1. All non-zero digits are significant. 198745 contains six significant digits.
2. All zeros that occur between any two non zero digits are significant.

For example, 108.0097 contains seven significant digits.

3. All zeros that are on the right of a decimal point and also to the left of a non-zero digit is never significant.

For example, 0.00798 contained three significant digits.

4. All zeros that are on the right of a decimal point are significant, only if, a non-zero digit does not follow them.

For example, 20.00 contains four significant digits.

5. All the zeros that are on the right of the last non-zero digit, after the decimal point, are significant.

For example, 0.0079800 contains five significant digits.

6. All the zeros that are on the right of the last non-zero digit are significant if they come from a measurement.

For example, 1090 m contains four significant digits.



Counting Significant Figures

The Digits	Digits That Count	Example	# of Sig Figs
Non-zero digits	ALL	<u>4.337</u>	4
Leading zeros (zeros at the BEGINNING)	NONE	0.000 <u>65</u>	2
Captive zeros (zeros BETWEEN non-zero digits)	ALL	<u>1.000023</u>	7
Trailing zeros (zeros at the END)	ONLY IF they follow a significant figure AND there is a decimal point in the number	<u>89.00</u> but <u>8900</u>	4 2
Leading, Captive AND Trailing Zeros	Combine the rules above	0.00 <u>3020</u> but <u>3020</u>	4 3
Scientific Notation	ALL	<u>7.78</u> x 10 ³	3

Accuracy versus precision

Accuracy

The ability of an instrument to measure the accurate value is known as accuracy. In other words, it is the ***closeness of the measured value to a standard or true value.***

Accuracy Formula

Accuracy is calculated using the following percent error formula:

$$\text{Percent Error} = \{(\text{Measured Value} - \text{True Value}) / \text{True Value}\} \times 100$$

This formula gives us the accuracy as a percentage.

The less the percent error the more accurate the value is.

Precision

- The closeness of two or more measurements to each other is known as the precision of a substance.
- If you weigh a given substance five times and get 3.2 kg each time, then your measurement is very precise but not necessarily accurate.
- Precision is independent of accuracy.

Precision is sometimes separated into:

1. Repeatability

The variation arising when the conditions are kept identical and repeated measurements are taken during a short time period.

2. Reproducibility

The variation arises using the same measurement process among different instruments and operators, and over longer time periods.

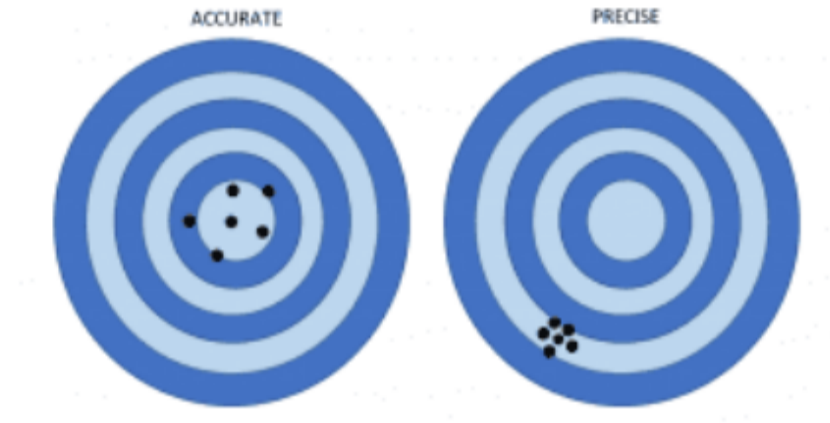
Examples

- ✓ If the weather temperature reads $28\text{ }^{\circ}\text{C}$ outside and it is $28\text{ }^{\circ}\text{C}$ outside, then the measurement is said to be accurate. If the thermometer continuously registers the same temperature for several days, the measurement is also precise.
- ✓ If you take the measurement of the mass of a body of 20 kg and you get 17.4 , 17 , 17.3 and 17.1 , your weighing scale is precise but not very accurate. If your scale gives you values of 19.8 , 20.5 , 21.0 , and 19.6 , it is more accurate than the first balance but not very precise.

Example of the difference between Accuracy and Precision...

The example of a darts board is often used when talking about the difference between accuracy and precision.

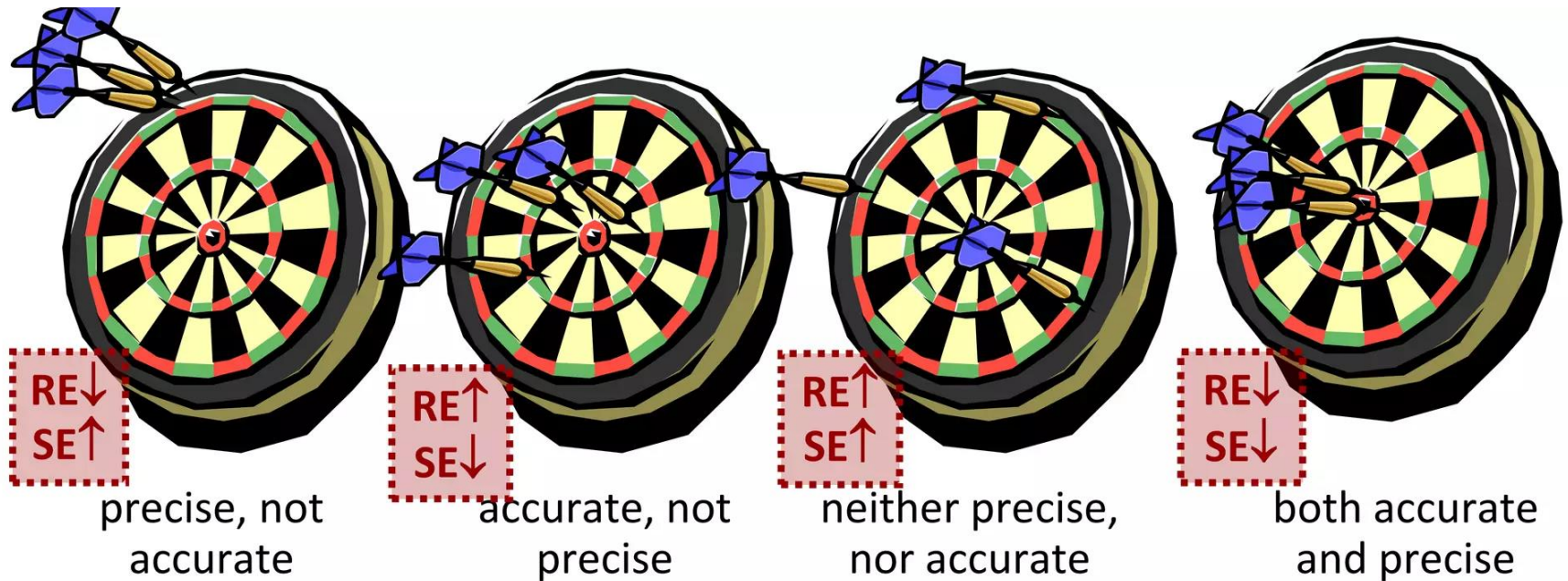
Accurately hitting the target means you are close to the centre of the target, even if all the marks are on different sides of the centre. Precisely hitting a target means all the hits are closely spaced, even if they are very far from the centre of the target.



Note

Accuracy is the degree of closeness between a measurement and its true value.

Precision is the degree to which repeated measurements under the same conditions show the same results.



- A measurement is said to be accurate if it has little systematic errors.
- A measurement is said to be precise if it has little random errors.
- A measurement can be of great precision but be inaccurate (for example, if the instrument used had a zero offset error).

Difference between Accuracy and Precision

Accuracy	Precision
Accuracy is closeness with the true value of the quantity being measured.	Precision is a measure of the reproducibility of the measurement.
Measurement can be accurate but not necessarily precise.	Measurement can be precise but not necessarily accurate.
It can be determined with a single measurement.	It needs several measurements to be determined.
Accuracy may be affected with systematic error.	Precision may be affected with random error.
Accurate values have to be precise in most cases.	Precise values may or may not be accurate.
Degree of conformity.	Degree of reproducibility.

InstrumentationTools.com

Error versus uncertainty

Error

An error is a “mistake” in a measurement.

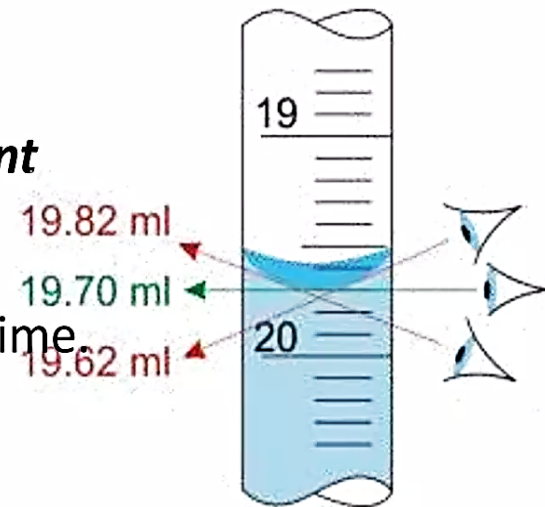
Types of errors:

- Random
- Systematic

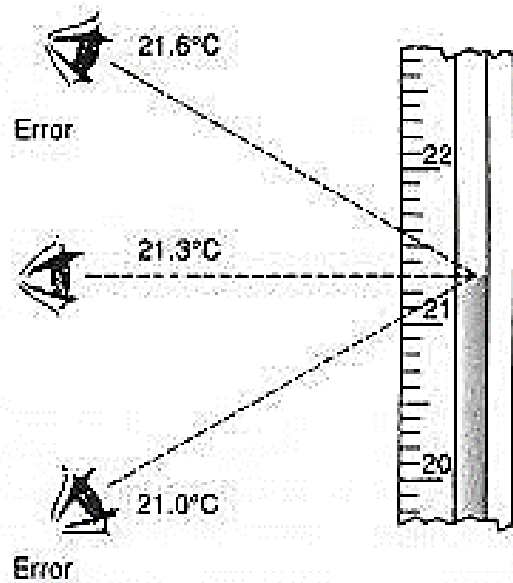
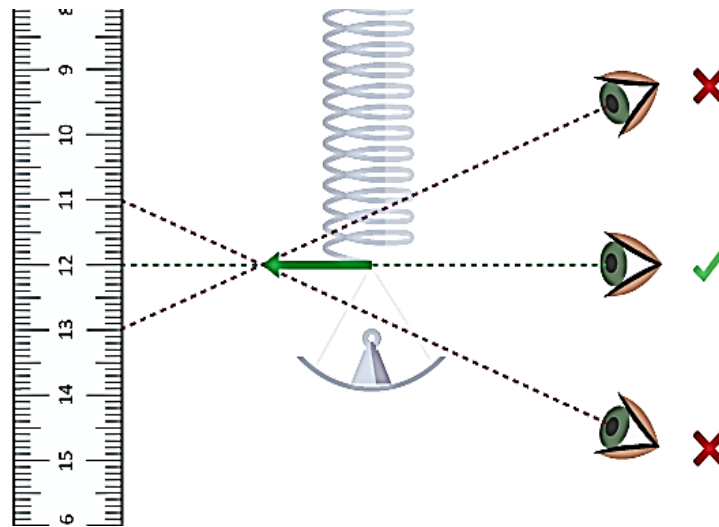
Random Errors

◆ **Random:** refer to random fluctuations in the measured data due to:

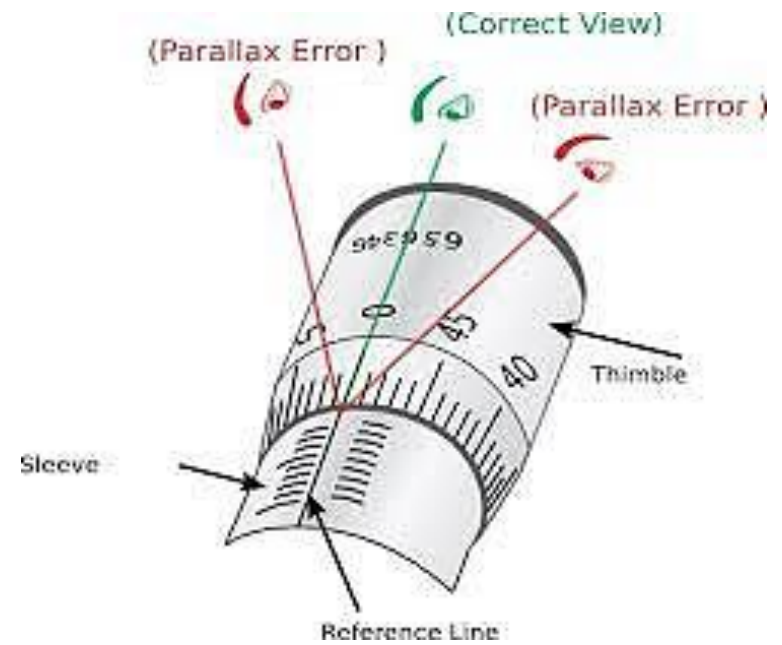
- the readability of the instrument
- the effects of something changing in the surroundings between measurements
- the observer being less than perfect 🪄
 - poor technique (e.g. carelessness with parallax)
The observer being less than perfect in the **different way** during each measurement.
 - Perhaps the ruler wasn't perfectly lined up every time



- Random errors can be reduced by averaging.
A precise experiment has small random error.



(a) Temperature measurement



Systematic errors

Systematic errors is error due to the instrument being “out of adjustment.”

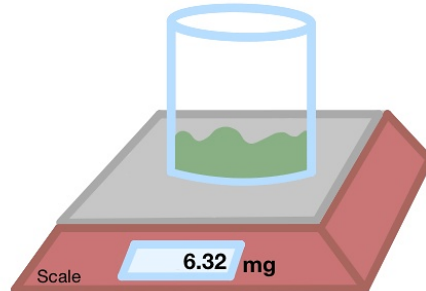
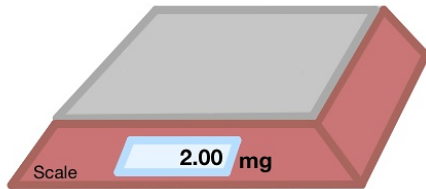
- An instrument with a zero offset error.
- A meter stick might be worn off or rounded at one end
- An instrument that is improperly calibrated
- poor technique (e.g. carelessness with parallax)

The observer being less than perfect in the ***same way*** every time.



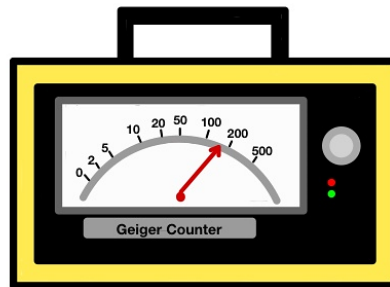
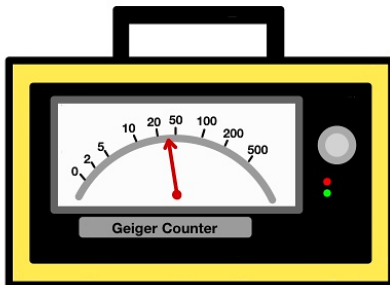
- Systematic errors are usually difficult to detect.
- Systematic errors can be detected using **different methods** of measurement

Systematic Error



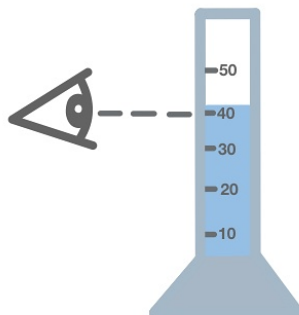
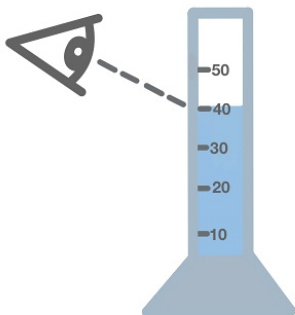
Instrumental Error

The scale is improperly calibrated so it reads 2mg even with nothing on it. This causes the the beaker to appear 2mg heavier than it would be, and any other measurements would read 2mg heavier than what they actually weigh



Environmental Error

The Geiger counter shows the radiation, however, when a cell phone is near the Geiger counter, the radiation is read as greater than what it actually is because the cell phone is radiating RF waves which disrupts the reading

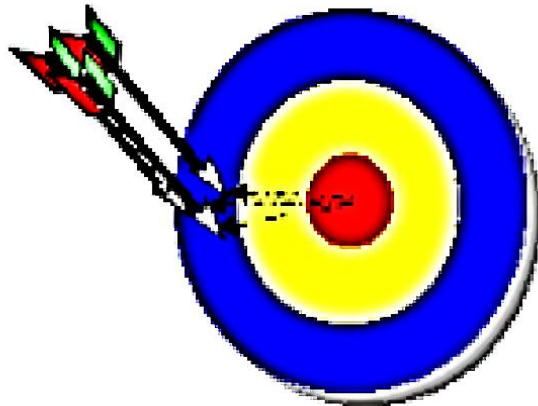


Observational Error

The observer looks at the beaker from above eye level and write down the volume as 40mL, when in reality when looked at eye level you can see the liquid has a volume of about 42mL

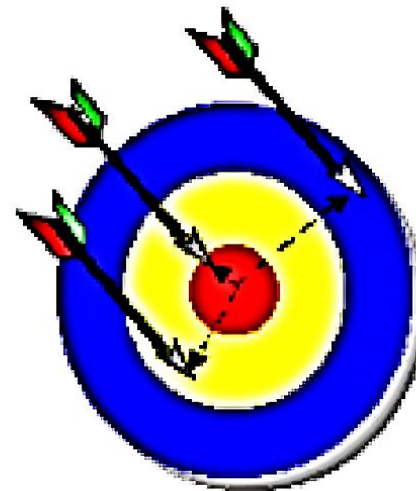
systematic error

- ❖ poor accuracy
- ❖ definite causes
- ❖ reproducible



random error

- ❖ poor precision
- ❖ nonspecific causes
- ❖ not reproducible



Absolute Error

Absolute error = | True value – Measured value |

Relative Error and Percent Error

$$\text{Relative Error} = \frac{|\text{measured} - \text{real}|}{\text{real}}$$

$$\text{Percent Error} = \frac{|\text{measured} - \text{real}|}{\text{real}} \times 100\%$$

Absolute, Relative, and Percent Error

$$\text{Absolute Error} = |\text{True Value} - \text{Measured Value}|$$

Absolute error is the magnitude of the difference between a measurement compared to a true value.



$$\text{Relative Error} = \text{Absolute Error} / \text{True Value}$$

Relative error shows the size of error relative to the true value.

Basically, it puts the size of the error into perspective.



$$\text{Percent Error} = \text{Relative Error} \times 100\%$$

Percent error shows what percent the error is of the true or accepted value.

UNCERTAINTIES

- ▶ **No measurement is perfect**
 - Our estimate of nearness to the true value is called the uncertainty (or error)
- ▶ **Uncertainty in data leads to uncertainty in calculated results**
 - Uncertainty never decreases with calculations, only with better measurements
- ▶ **Reporting uncertainty is essential**
 - The uncertainty is critical to decision-making
 - Estimating uncertainty is your responsibility

Uncertainties

Every measurement has uncertainty.

Why?

- **The measuring device**
- **Experimental technique**
- **Nature of the measurement itself**
 - **For example the weight of a grain of sand**

Measurement Uncertainty

Measurement uncertainty is the result of imprecision in measuring devices. All measuring devices have some degree of imprecision and uncertainty.

Measurement uncertainty is always the larger of:

1: The limit of the instrument (half the smallest increment) multiplied by the number of measurements taken.

2: A justified estimation of the limit of the measurement procedure as done by the experimenter.

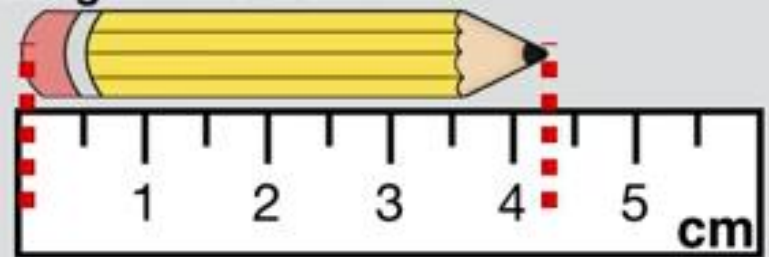
1: The limit of the instrument (half the smallest increment) multiplied by the number of measurements taken.

Smallest increment = 0.5 cm

of measurements = 2

Uncertainty = $2 \times 0.25 \text{ cm} = 0.5 \text{ cm}$

Length = $4.3 \pm 0.5 \text{ cm}$



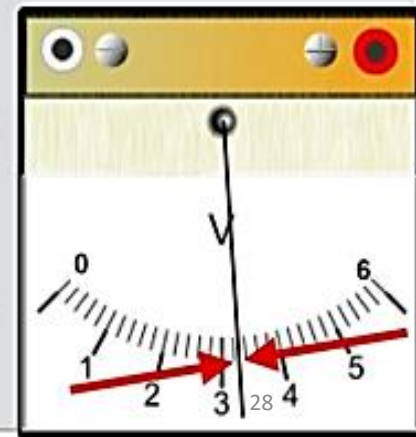
measurements taken.

Smallest increment
= 0.2 V

of measurements = 2

Uncertainty = $2 \times 0.1 \text{ V}$
= 0.2 V

Voltage = $3.3 \pm 0.2 \text{ V}$



How to calculate the uncertainty?

Repeated Measurements

- When you repeat the measurement many times.
- Find the average of all the trials
- Then:
 - Uncertainty = $\pm 1/2$ (Max – min)

Repeat Measurements Example

Rama measures the height of a chair 5 times. He gets the following results:

2500 mm, 2504 mm, 2509 mm, 2506 mm and 2501 mm



Average value = 2504 mm

Uncertainty = $(2509 - 2500)/2 = 4.5$ mm

Height of the chair = 2504 ± 5 mm

This means the true height is between 2499 and 2509 mm

Absolute uncertainty

- Absolute uncertainty is the uncertainty you get from a measurement.

From the example earlier, the height of the chair was found to be 2504 ± 5 mm

The **absolute** uncertainty is **5 mm**

- remember: some instruments give a predetermined uncertainty which you **must** use.

Fractional uncertainty

- Fractional uncertainty is calculated by dividing the absolute uncertainty by the measured value

From the example earlier, the height of the chair was found to be 2504 ± 5 mm

The **fractional** uncertainty is **$5/2504 = 0.002$**

Calculating Percent Uncertainty

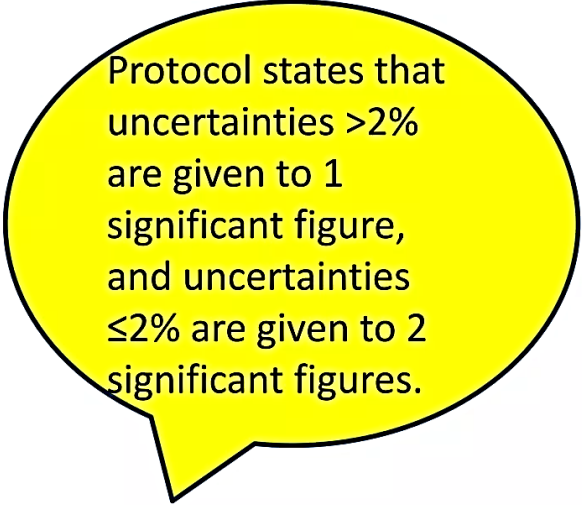
The % uncertainty is calculated by dividing the absolute uncertainty by the measured value and the multiplying by 100.

$$\frac{\text{Absolute uncertainty}}{\text{measured value}} \times 100 = \text{Percent Uncertainty}$$

Calculating Percent Uncertainty

From the example earlier, the height of the chair was found to be 2504 ± 5 mm

The **percentage** uncertainty is
 $5/2504 \times 100 = 0.20\%$



Protocol states that uncertainties $>2\%$ are given to 1 significant figure, and uncertainties $\leq 2\%$ are given to 2 significant figures.

Absolute, Fractional, and Percent Uncertainty

Absolute Uncertainty	Fractional Uncertainty	Percent Uncertainty
Uncertainty expressed as a number independent of the original number Symbol: Δa or Δb or Δc etc. Example: 4 ± 1 Convert from absolute to fractional uncertainty: Divide the absolute uncertainty by the number $\frac{1}{4} = 0.25$ Convert from fractional to absolute uncertainty: Multiply the fraction by the number $0.25 \times 4 = 1$	Uncertainty expressed as a fraction of the original number Symbol (and equation): $\frac{\Delta a}{a}$ or $\frac{\Delta b}{b}$ or $\frac{\Delta c}{c}$ etc. Example: 4 ± 0.25 Convert from fractional to percent uncertainty: Multiply the fractional uncertainty by 100% $0.25 \times 100\% = 25\%$ Convert from percent to fractional uncertainty: Divide the fractional uncertainty by 100% $\frac{25\%}{100\%} = 0.25$	Uncertainty expressed as a percentage of the original number. Symbol (and equation): $\frac{\Delta a}{a} \times 100\%$ or $\frac{\Delta b}{b} \times 100\%$ or $\frac{\Delta c}{c} \times 100\%$ etc. Example: $4 \pm 25\%$

Calculating Percent Uncertainty

Example: Your turn to try 😊

	Measurement	Absolute Uncertainty	Percent Uncertainty
Density	1.15 g/cm ³	±0.05 g/cm ³	$\frac{0.05 \text{ g/cm}^3}{1.15 \text{ g/cm}^3} \times 100 = 4\%$
Weight	59.67 g	±0.01 g	$\frac{0.01 \text{ g}}{59.67 \text{ g}} \times 100 = 0.017 \%$

Rule 1: Addition/Subtraction

- When you are adding or subtracting values in a calculation, the uncertainty in the calculated answer is equal to the sum of the absolute uncertainties for each measurement added:

$$\begin{array}{r} 24.10 \pm 0.05 \text{ g} \\ + 13.05 \pm 0.02 \text{ g} \\ \hline = 37.15 \pm 0.07 \text{ g} \end{array}$$

2. Rule for multiplication and division

The fractional uncertainty of a product or quotient is the sum of the fractional uncertainty of the operands: (??)

$$\text{If } y = \frac{ab}{c}, \text{ then } \frac{\Delta y}{y} = \frac{\Delta a}{a} + \frac{\Delta b}{b} + \frac{\Delta c}{c}$$

Δ - uncertainty in...

Simple Rules

▶ **Rule 1:**

If a measured quantity is multiplied or divided by a constant then the *absolute* uncertainty is multiplied or divided by the same constant. (In other words the relative uncertainty stays the same.)

▶ **Example to illustrate rule 1**

Suppose that you want to find the average thickness of a page of a book. We might find that 100 pages of the book have a total thickness of 9mm. If this measurement is made using an instrument having a precision of **0.1mm**, we can write

Thickness of 100 pages, $T = 9.0\text{mm} \pm 0.1\text{mm}$

and, the average thickness of one page, t , is obviously given by

$$t = T/100$$

therefore our result can be stated as $t = 9/100\text{mm} \pm 0.1/100\text{mm}$
or

$$\mathbf{t = 0.090\text{mm} \pm 0.001\text{mm}}$$

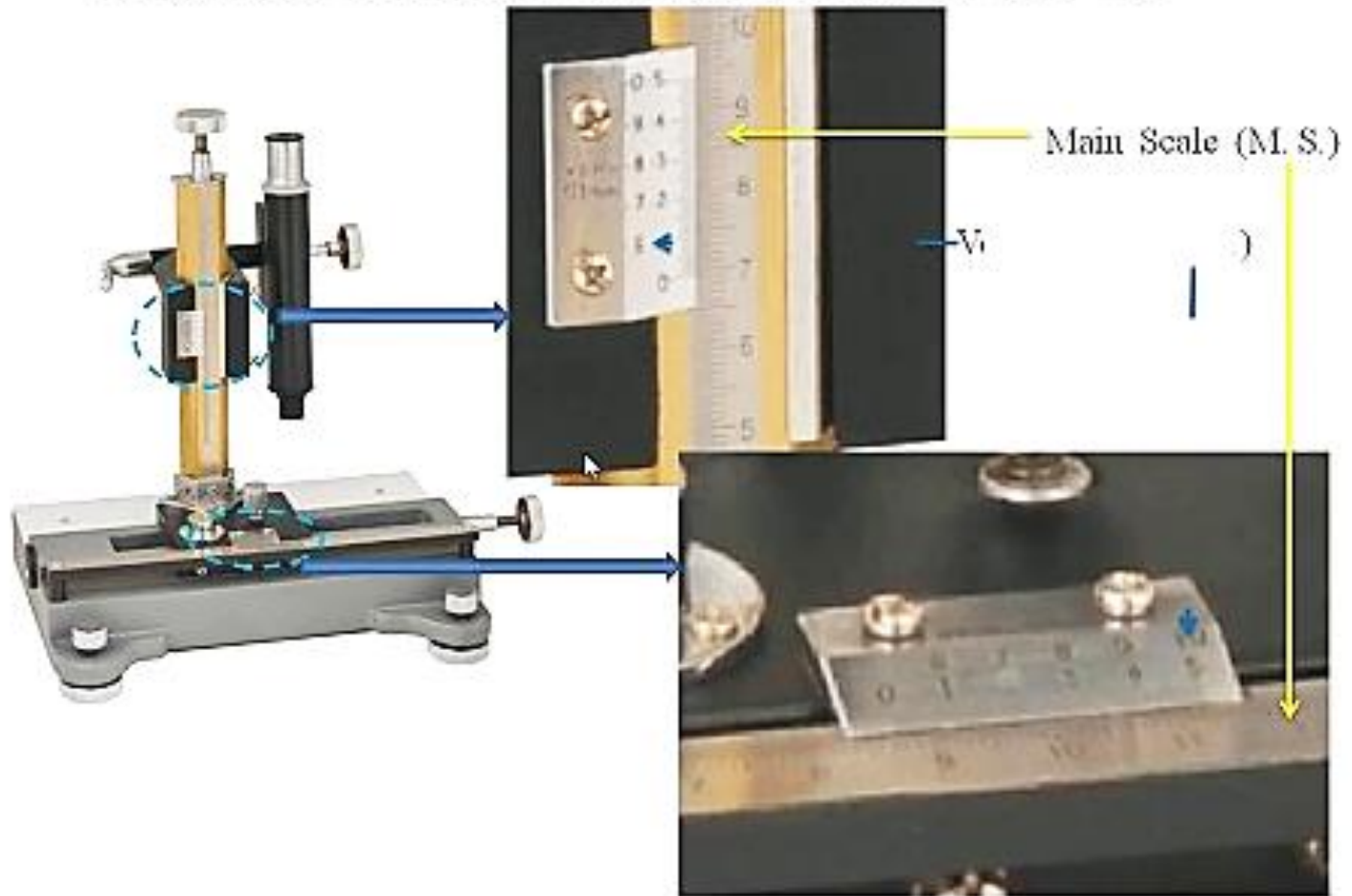
Least Count

Least count of a measuring instrument is the smallest measurement that can be made accurately with the given measuring instrument.

A meter scale normally has graduations at 1 mm (or 0.1 cm) spacing, as the smallest division on its scale. You cannot measure lengths with this scale with accuracy better than 1mm (or 0.1 cm).

$$L.C = \frac{\text{value of 1 main scale division}}{\text{Total no of division on vernier Scale}}$$

TRAVELLING MICROSCOPE



A travelling microscope is a precision instrument used for measuring small distances and angles with high accuracy. It is a type of compound microscope that is designed to be portable and easy to use.

Travelling microscopes are commonly used in scientific research, engineering, and manufacturing. They are particularly useful for measuring the thickness of thin films, the diameter of small objects, and the distance between two points on a surface.



M.S.R :- 11 DIVISIONS AFTER 0 AND ONE M.S.D = 0.05 cm

M.S.R :- $11 \times 0.05 = 0.55$ cm

V.S.D :- 44th DIVISION IS BEST COINCIDING WITH MAIN SCALE

P :- $M.S.R + V.S.D \times L.C$

:- $0.55 + 44 \times 0.001 = 0.594$ cm

0.6 0.7 0.8 0.9 1.0
0 1 2 3 4 5

Least count of a Travelling Microscope

C

0.01 mm

SOLUTION

$$1 \text{ MSD} = 0.5 \text{ mm}$$

$$50 \text{ VSD} = 49 \text{ MSD}$$

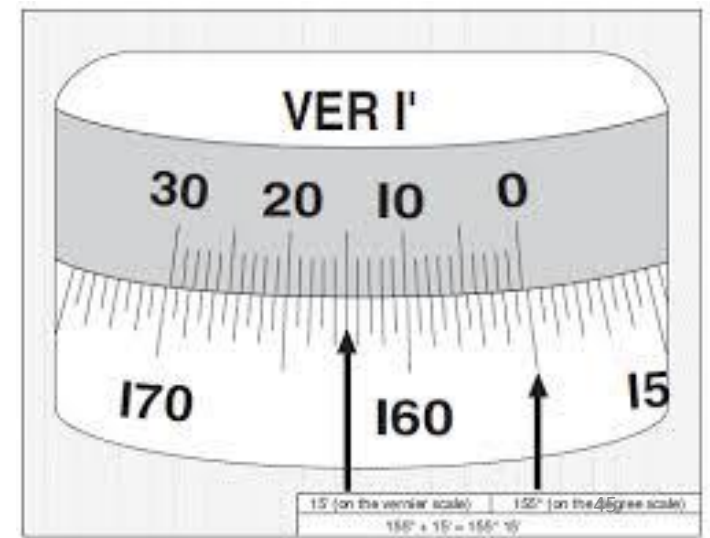
$$1 \text{ VSD} = \frac{49}{50} \text{ MSD}$$

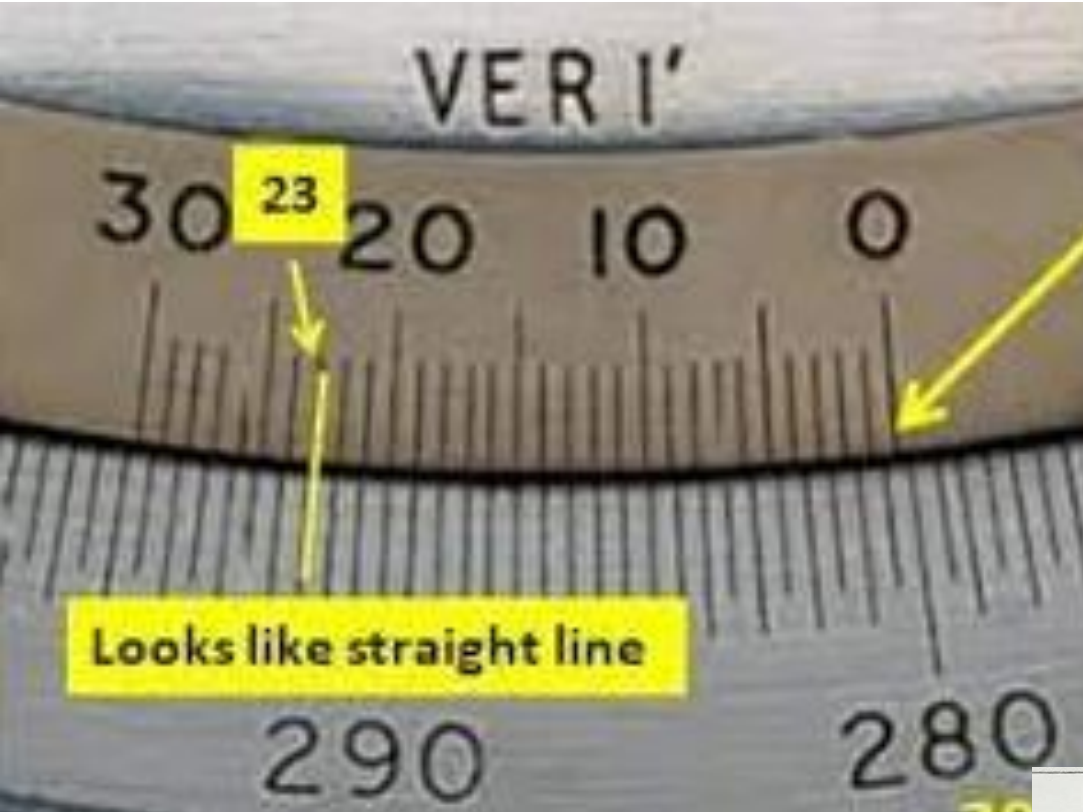
$$\text{Least count} = 1 \text{ MSD} - 1 \text{ VSD}$$

$$= 1 \text{ MSD} - \frac{49}{50} \text{ MSD} = \frac{1}{50} \text{ MSD} = \frac{1}{50} \times 0.5 \text{ mm} = 0.01 \text{ mm}$$

Least count of a spectrometer

A spectrometer is a scientific instrument used to separate and measure spectral components of a physical phenomenon.





LC of a Spectrometer.

$$LC = \frac{\text{Value of 1 M.S.D}}{\text{Tot. No. of divisions on V.S.}}$$

$$10^\circ \rightarrow 30 \text{ div.}$$

$$1^\circ \rightarrow 3 \text{ div}$$

$$1 \text{ div} = \frac{1^\circ}{3}$$

$$\therefore LC = \frac{\frac{1^\circ}{3}}{30} = \left(\frac{1}{90}\right)^\circ$$

Thank you!